

Targeted Model Repair under a Hidden Clean Teacher

A competition study of Neural Debris Removal in RetinaNet streak detection

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github.com/gtom-pandas/neural-debris-hermes-research

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Abstract

The Neural Debris Removal in Streak Detection Models competition asked participants to repair a poisoned RetinaNet detector for 1024 x 1024, 16-bit night-sky images. The clean model and all test labels were hidden; only the poisoned checkpoint, 20 annotated poison examples, 2,000 test images, and the poisoned model's submission template were provided. The evaluation metric, mean asymmetric Confidence-Aware Detection Distance (maCADD), penalized confidence and localization disagreement with the hidden clean model while discounting confidence changes in the intended unlearning direction.

This report reconstructs our complete competition trajectory from the local Git history, 40 authenticated Kaggle kernels, 19 leaderboard submissions, stored run audits, and the post-competition public solutions. We progressed from a catastrophic repair score of 449.9513 through a stable 257-260 basin, then to 245-248 with public pruning and fine-tuning reproductions, 229.1934 with a pruning plus short EWC fine-tune, and a best public score of 219.4257 using a conservative confidence-selection policy. The best private score was 306.3502, producing private rank 89; the public-best candidate scored 308.0971 privately. The winner scored 178.6560 privately. This large residual gap and the weaker private generalization show that public-score optimization outpaced reliable identification of clean detections.

Our central finding is that the project identified many correct ingredients - controlled unlearning, preservation losses, distillation, geometry audits, synthetic pairs, and confidence calibration - but did not integrate them into a sufficiently dense, anchor-level teacher-student objective with concept-level synthetic validation. The leading published solution did exactly that. We report both successful and abandoned directions, separate observations from causal claims, and conclude with the missing experiments and repository changes required for a reproducible model-repair study.

Keywords: machine unlearning, backdoor removal, object detection, RetinaNet, Detectron2, knowledge distillation, EWC, confidence calibration, Kaggle, space debris.

1. Introduction

Optical surveys detect artificial objects in Earth orbit as faint streaks against a noisy night-sky background. A detector intended for operational use must be accurate, but it must also remain robust when part of its training data has been manipulated. The competition turned that security problem into a controlled model-repair task: one clean RetinaNet was kept hidden, a poisoned version was distributed, and competitors had to update the poisoned weights so that its detections approached the clean model while removing the backdoor behavior [1].

This differs from standard supervised object detection in three ways. First, there is no clean training set and no clean validation labels. Second, the 20 available examples show what must be forgotten, not what must be retained. Third, the metric compares an unknown set of clean detections with a submitted set, so every extra box and every confidence error has a cost. Empty-label fine-tuning can suppress poison, but without a preservation signal it can also erase real streaks or create new false detections.

Our work was conducted as an independent Kaggle study supported by an agentic research workflow: a local Codex workspace, a VPS-hosted Hermes agent, Kaggle CLI/API access, and machine-readable run audits. The aim was not only to improve the leaderboard score but to make each action reviewable and bounded. Every submitted file was tied to a named hypothesis; most candidate runs remained no-submit experiments.

The contributions of this report are:

- a verified reconstruction of the challenge, metric, data, model, and submission history;
- a chronological account of the methods tried, including failed and rejected families;
- a distinction between the public-best score, the private-best score, and the final rank;
- a comparison with the leading published teacher-student solutions;
- a concrete diagnosis of what was missing technically and reproductively.

2. Challenge formulation

2.1 Data and detector

The organizers supplied a Detectron2 RetinaNet checkpoint and a public unlearn set of 20 1024 x 1024 grayscale PNG images. Each image contained one annotated poisoned streak. The test set contained 2,000 unlabeled images. Our static checkpoint audit found 301 tensors and approximately 36.36 million parameters or buffers. The head shapes were consistent with a single-class RetinaNet using seven anchors per location: classification weights of shape 7 x 256 x 3 x 3 and box-regression weights of shape 28 x 256 x 3 x 3.

The official sample submission contained 2,000 rows and encoded detections as repeated groups of five values: confidence, x, y, width, and height. The poisoned-model sample had 2,593 detections, a median of one detection per image, and up to eight detections on a single image. Reproducing the organizer inference path required the non-default anchor aspect ratios 0.1, 0.2, 0.5, 1, 2, 5, and 10.

2.2 Mean asymmetric Confidence-Aware Detection Distance

At each IoU threshold t in $\{0.2, 0.3, \dots, 0.9\}$, detections are matched and confidence differences are accumulated. Unmatched clean detections contribute their full confidence; unmatched submitted detections contribute their full submitted confidence. For matched pairs, a confidence change in the correct unlearning direction receives a factor $1/A$, where $A = 10$, while a change in the wrong direction is charged at full weight. The final metric is the IoU-weighted mean:

$$\text{maCADD} = \sum(t * \text{aCADD}_t) / \sum(t), \text{ for } t \text{ in } \{0.2, 0.3, \dots, 0.9\}.$$

Lower is better and zero is perfect. The clean reference includes only detections with confidence above 0.2. The asymmetry makes confidence calibration part of the solution, but it does not make extra detections free: an unmatched box is still charged at its submitted confidence. This tension explains several of our regressions.

2.3 Evaluation constraints

The rules prohibited manual, weak, soft, or pseudo-label annotation of the test set for competitive advantage. Predictions from the supplied poisoned model could be analyzed, and the unlabeled test set could be used as reference data during de-poisoning. The daily limit was two submissions per team during most of our work. The final ranking used the private split.

3. Evidence and audit protocol

This report is based on five independent evidence surfaces: the Git history, the sanitized public repository, Kaggle competition pages retrieved through the authenticated API, the account's 40 kernels and their output audits, and the 19 scored submissions. The local worktree was clean before the audit, and the public GitHub main branch matched the local commit bed2805.

For every available candidate, we preferred stored facts over retrospective inference: checkpoint hashes, row counts, number of detections, empty-image counts, confidence mass, score distributions, and public/private leaderboard values. A method is called successful only when its submitted score improved the relevant baseline. A no-submit run can support a negative conclusion, but it is not assigned an imagined leaderboard score.

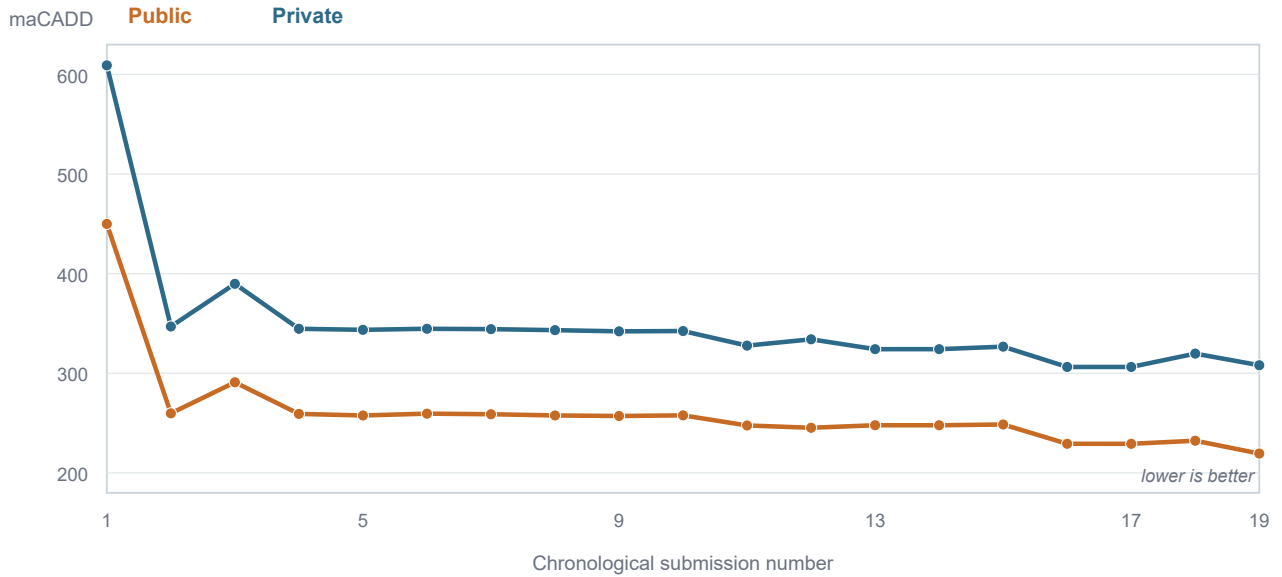


Figure 1. All 19 account submissions in chronological order. Lower maCADD is better. The private split is consistently harder, and the final public gains do not close the private gap.

4. Research trajectory

4.1 Contract reconstruction and the first collapse

The first phase reconstructed Detectron2 inference, audited the checkpoint, and verified the output contract. The initial repair-suppress-001 attempt scored 449.9513 public and 609.1727 private. This established that aggressive suppression without reliable retention was destructive. A simple empty-label fine-tuning baseline improved to 259.7886 public and 346.9767 private, while an over-trained variant regressed to 290.9630/389.8185. The iteration count was therefore not a minor hyperparameter: the repair trajectory crossed a narrow useful region before collapsing useful behavior.

4.2 Interpolation and conservative preservation

We next treated the useful MAX_ITER=20 checkpoint as an anchor and interpolated it with nearby checkpoints. A 5% interpolation toward MAX_ITER=30 reached 257.6416 public, followed by 257.6576 at 6%; other local mixtures remained near 258-259. This was a real but small improvement. The family changed confidence mass and empty-image counts smoothly, yet it did not identify poison-specific behavior. We therefore abandoned interpolation as a primary route rather than spending the submission budget on a finer grid.

Controlled-unlearning kernels added teacher-confidence preservation, outside-poison losses, L2-SP regularization, ranking preservation, and a narrow confidence corridor. The softest submitted variant reached 257.1246 public. These experiments anticipated the correct preserve/forget decomposition, but the implementation operated on a restricted repair scope and local corridor targets rather than a complete anchor-level distillation loss for classification and box deltas.

4.3 Public reproductions: pruning, noise, and EWC

Public notebooks provided stronger hypotheses. A reproduced pruning plus EWC recipe reached 247.8306 public. A separate public de-poisoning notebook reached 245.3014, but its private score was 334.1291, worse than several later candidates. Re-submitting the same Jayhawk CSV reproduced 247.8306 exactly, which validated the pipeline but not the method's competitiveness.

The Jayhawk family used activation-guided channel pruning, calibrated noise, a short fine-tune, and EWC. We swept EWC strength, pruning fraction, noise amplitude, seeds, and operation order. A representative reversed-order run (noise then 15% pruning, EWC 4000, 50 iterations, learning rate 1e-4) produced only 1,452 detections and 963 empty images. It was not promoted. The lesson was that pruning/noise perturbed a circuit shared by poison and real streaks; ordering and magnitude changed density but did not create a robust poison-only separator.

4.4 Causal synthetic pairs and restricted model repair

The causal-synthetic-pairs-unlearn-001 kernel generated original, masked, moved, and perturbed poison patches. It trained 3.29 million parameters across selected FPN and classification layers for 80 iterations with learning rate 5e-5 and an outside-patch consistency term. The resulting submission profile had 2,180 detections, 714 empty images, and confidence mass 1010.82. This returned to the known 257-style basin and was rejected before submission.

This was an important negative result. Synthetic intervention alone was not enough. The pair construction lacked an equally strong, teacher-pinned clean-streak objective, so the network could preserve broad features while still failing to separate poison from the faint end of the real-streak distribution.

4.5 The Sanidhya branch and the 229 basin

A later exact reproduction of a public Sanidhya pipeline improved the public score to 229.2314. We then separated its pruning-only and mixed short-fine-tune stages. The selected Phase B candidate combined pruning with a short EWC fine-tune (EWC 250, 10 iterations), producing 2,582 boxes: 1,440 at epsilon confidence, 634 empty images, and total confidence mass 295.56. It scored 229.1934 public and 306.3502 private, the best private score on the account.

The fixed geometry enabled a family of no-submit oracle and remapping studies. Independent public checkpoints were re-inferred at lower thresholds; selected candidates from Alok and Include4eto passed mechanical review, while DataArthur, Ruhul, Mirza, and a reset classification head were rejected for excessive density, confidence mass, weak geometry agreement, or complete collapse. Pixel-level dashedness, erase-persistence, and multi-oracle disagreement changed only small subsets of scores and did not justify a leaderboard slot.

4.6 Metric-aware rescoring and the final public best

The Amer reproduction treated the frozen poisoned model as a candidate generator at confidence 0.05. It combined survival under four classifier-only unlearners, poison-box geometry, a crop appearance classifier, and transplanted poison patches. Its conservative output preserved organizer geometry while reducing suspicious candidates to epsilon. This formalized a useful insight: when box geometry is already close to the hidden clean model, the remaining problem is candidate selection and confidence estimation.

The ndr-trial-v2 kernel extended this into a contrastive weight-space ranking problem. One model acted as an unlearner; a second poison amplifier moved in the opposite direction. The planned selector ranked candidates by high survival under unlearning, low amplification, weak poison-prototype evidence, and original confidence. Its submitted center variant scored 232.3060 public and 319.7617 private, failing to beat Phase B.

The final public-best notebook did not use its title's direct external fine-tune as the scored method; that direct route had scored about 260.31 and was rejected. Instead, it reconstructed a previously selected global mask over the organizer template. It retained at most two candidates per image and 450 globally, assigned confidence 0.02 to all non-selected candidates, and added 0.15 to the organizer confidence of selected candidates. The result scored 219.4257 public but 308.0971 private. The public gain was genuine on the public split, yet the private result was slightly worse than Phase B. This is the clearest evidence that the selector had learned leaderboard-specific decision boundaries rather than a sufficiently stable approximation of the hidden clean detector.

5. Results

5.1 Leaderboard progression

Date	Candidate	Public	Private	Decision
Jun 08	repair-suppress-001	449.9513	609.1727	Catastrophic; abandon

Date	Candidate	Public	Private	Decision
Jun 11	simple fine-tune, useful region	259.7886	346.9767	Baseline
Jun 11	simple fine-tune, over-trained	290.9630	389.8185	Reject
Jun 18	interpolation beta 0.05	257.6416	343.6334	Small gain; plateau
Jun 21	preserve-soft unlearning	257.1246	342.1067	Stable but weak
Jun 23	public CV de-poisoning	245.3014	334.1291	Public gain, poor private
Jun 24	Jayhawk pruning + EWC	247.8306	324.2157	Reproducible
Jun 28	noise 0.03	248.5938	326.7423	No gain
Jul 06	Sanidhya reproduction	229.2314	306.3586	New basin
Jul 18	Sanidhya Phase B	229.1934	306.3502	Best private
Jul 21	contrastive trial v2	232.3060	319.7617	Reject
Jul 23	final confidence selector	219.4257	308.0971	Best public only

The public and private scores are strongly correlated across the account's submissions, but this does not imply that the public-best candidate is the private-best candidate. The final selector improved public maCADD by 9.77 points over Phase B and worsened private maCADD by 1.75 points.

5.2 Final standing and top-leaderboard gap

The exported public leaderboard placed the account at rank 78 with 219.4257. The released private leaderboard placed team T Soo at rank 89 with 306.3502. Kaggle's current competition metadata reports 567 teams after organizer validation and removal of duplicate or cooperating accounts.

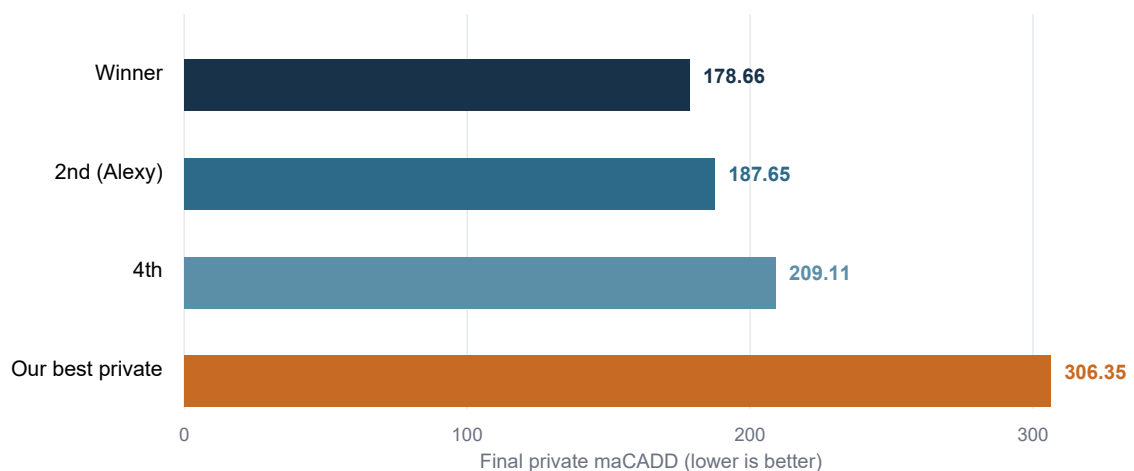


Figure 2. Final private maCADD for the winner, the published second- and fourth-place solutions, and our final private-best entry. The 127.69-point gap to the winner is much larger than the final public gap.

5.3 What worked and what did not

Family	Evidence	Outcome
Empty-label fine-tuning	259.79 useful, 290.96 over-trained	Narrow useful region; high collateral damage
Checkpoint interpolation	Best 257.64	Smooth local improvement, no poison-specific separation
Preserve-aware losses	Best 257.12	Correct principle, incomplete dense implementation
Pruning/noise/EWC	247.83; many no-submit sweeps	Better public basin, unstable density/retention
Public CV reproduction	245.30 public, 334.13 private	Strong public signal, weak private generalization
Causal synthetic pairs	No-submit; returned to 257 profile	Rejected; insufficient clean pinning
Sanidhya mixed Phase B	229.19 public, 306.35 private	Best private result
Multi-oracle remaps	Mostly rejected locally	Signals too correlated or too weak
Contrastive selector	232.31/319.76	Promising formulation, inferior selection precision
Final confidence mask	219.43/308.10	Best public, mild private regression

6. Comparison with leading solutions

As of August 1, the winner had not published a detailed solution. The final private score was 178.6560. The second-place solution by Alexy reported 143.6582 public and 187.6500 private, while the fourth-place solution described a related knowledge-distillation constrained unlearning method and scored 209.1104 privately [8,9].

The second-place method is the most informative comparison because it documented its full experiment log. It froze the poisoned detector as a teacher and trained a student with two explicit objectives. Outside the poison-anchor mask, student classification logits and box deltas were pinned to the teacher. Inside the mask, classification logits above a soft target corresponding to confidence 0.03 were suppressed. The poison mask used intersection-over-anchor rather than ordinary IoU so that large anchors containing a short poison streak were not missed.

The decisive step was concept-level generalization. The authors pasted rotated poison patches and rendered bright beaded or comet-like streaks as poison examples, while smooth synthetic streaks served as distillation-only positives. Held-out synthetic poison detections fell from 26/31 to 5/31. A mild uniform confidence boost then improved public maCADD from 146.13 to 143.95, and a three-model consensus with per-detection confidence restoration reached 143.66 public and 187.65 private.

Our trajectory overlapped with this solution in vocabulary but not in integration. We used teacher confidence, preserve masks, ranking corridors, synthetic patches, pruning, EWC, and multi-model evidence. However, the components lived in separate kernels and were often used to rank or remap final detections. The leading solution made preservation the training objective itself, at every relevant anchor and box delta, then validated generalization on held-out synthetic concepts before calibrating the output.

7. What was missing relative to the best leaderboard

7.1 Dense preservation at the right representation level

Our public `train.py` learns a confidence repair head from exported detection-level features. Several kernels preserve aggregate confidence or selected logits. The best published method preserved all non-poison anchor logits and box deltas against a frozen teacher. This is a much stronger constraint: it prevents new boxes, box drift, and broad confidence collapse by construction.

7.2 A poison mask aligned with RetinaNet anchors

IoU between a short streak and a much larger anchor can be small even when the anchor fully contains the poison. Intersection-over-anchor better identifies the classification units that express the backdoor. Our experiments concentrated on output boxes, crop appearance, or selected feature channels; they did not consistently supervise this anchor-level causal boundary.

7.3 Concept-level synthetic data with explicit clean contrast

The 20 images are too small a sample to define a backdoor concept. Our causal-pair run moved and perturbed poison patches, but its clean contrast was not strong enough. The leading method used two clearly separated synthetic families: bright sparse poison-like patterns to forget and smooth real-like streaks to preserve. Faint ambiguous synthetic poison caused a documented collapse and was deliberately abandoned.

7.4 Label-free validation that measures both sides

We audited density, empty images, confidence mass, spatial agreement, and feature drift. Those checks catch collapse but do not directly prove concept selectivity. A stronger local suite would include held-out poison firing rate, confidence on long streaks that cannot be poison, new-detection counts against the teacher, box IoU retention, and synthetic clean retention. The best solution used these diagnostics as gates before spending submissions.

7.5 Confidence estimation after set selection

Our final gain came from choosing 450 boxes and boosting them. The public/private reversal shows that selection precision was insufficient. The leading solution first formed a stable consensus detection set, then restored confidence per detection from the poisoned teacher only when multiple repaired models supported that detection. The order matters: calibration cannot rescue a contaminated detection set.

7.6 Genuine model diversity

Many of our oracles were descendants of the same public recipes or evaluated the same fixed geometry. Their errors were therefore correlated. A useful ensemble needs distinct causal error modes, not only different thresholds, seeds, or confidence maps.

8. Repository and reproducibility audit

The GitHub repository is public, compact, and synchronized, but its public history stops on June 28. It documents the project as active with a best observed score of 245.3014. The actual competition record continued for another 25 days, added numerous probes and reached 219.4257 public. This gap makes the current repository unsuitable as the sole source for a final research claim.

The two public scripts are readable reference components. `train.py` trains a compact confidence repair head over exported detection features with poison, keep, and teacher signals. `infer.py` ensembles repair checkpoints, clips outputs, and writes a guarded CSV. They do not reproduce the full Detectron2 training path, the final selected mask, or the best private method. No automated tests are present.

For publication, the repository should contain a sanitized experiment registry for all 40 kernels, exact Kaggle version identifiers, hashes of non-redistributed inputs and outputs, contract tests, a full inference reproduction against the organizer sample, and a clear archival status. The raw datasets, secrets, and private operational logs should remain excluded.

9. Limitations

This study has no access to the hidden clean labels or model, so local causal attribution is necessarily limited. Leaderboard differences combine model repair, candidate selection, confidence calibration, and split variance. Several late experiments were selected using public feedback and therefore cannot be interpreted as unbiased estimates of generalization.

The top comparison is also incomplete. The winner had not released a detailed method by the audit date. We compare primarily with the documented second- and fourth-place solutions, and we do not claim that their mechanisms explain the winner's result.

Finally, the public repository was intentionally sanitized. Some raw run artifacts and operational context were removed from Git, although many remain recoverable through the authenticated Kaggle account. The reconstructed chronology is evidence-backed, but not every no-submit kernel has a common local metric or directly comparable output.

10. Conclusion

The Neural Debris project moved from indiscriminate suppression to a disciplined search for poison-specific repair. The score progression - 449.95, 259.79, 257.12, 245.30, 229.19, and finally 219.43 public - reflects real learning about collateral damage, confidence mass, geometry, and the limits of public recipes. The final private result of 306.35 and rank 89, however, shows that the public selector did not approximate the hidden clean detector well enough.

The most important technical lesson is that unlearning and preservation cannot be staged as loosely connected post-processing ideas. They must be encoded jointly at the detector's anchor and box-regression level, generalized with a clean/poison synthetic contrast, and validated on held-out concepts before confidence calibration. The most important research operations lesson is equally direct: a clean public repository is not a complete research record unless the final month, negative results, private score, and reproducibility boundary are consolidated after the competition.

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Appendix A. Complete submission history

#	Date	Description	Public	Private
1	Jun 08	repair-suppress-001	449.9513	609.1727
2	Jun 11	simple fine-tune baseline	259.7886	346.9767
3	Jun 11	simple fine-tune MAX_ITER=15	290.9630	389.8185
4	Jun 17	r02 cluster probe	259.2333	344.6891
5	Jun 18	interpolation beta 0.05	257.6416	343.6334
6	Jun 18	interpolation beta 0.02	259.4834	344.7010
7	Jun 19	triangular interpolation	258.8906	344.3229
8	Jun 19	interpolation beta 0.06	257.6576	343.3147
9	Jun 21	preserve-soft B1	257.1246	342.1067
10	Jun 22	fusion consensus average	257.7674	342.4261
11	Jun 23	public pruning + fine-tuning	247.6207	327.7771
12	Jun 23	public CV de-poisoning	245.3014	334.1291

#	Date	Description	Public	Private
13	Jun 24	Jayhawk reproduction	247.8306	324.2157
14	Jun 26	Jayhawk diagnostic re-submit	247.8306	324.2157
15	Jun 28	noise 0.03	248.5938	326.7423
16	Jul 06	NDR trial 1 / Sanidhya	229.2314	306.3586
17	Jul 18	Sanidhya Phase B	229.1934	306.3502
18	Jul 21	NDR trial v2	232.3060	319.7617
19	Jul 23	final confidence selector	219.4257	308.0971